

Segment - 2

"5BM" - 53

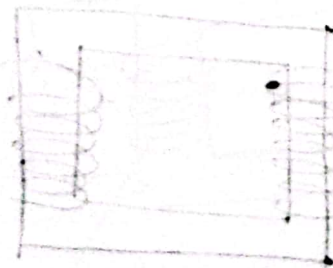
Transformer

Transformer is an electrical device that changes

the magnitude of alternating voltage or current from one value to another. It is a static device because it has no moving parts. Because of very rugged and durable in construction it requires very little maintenance.

Uses:

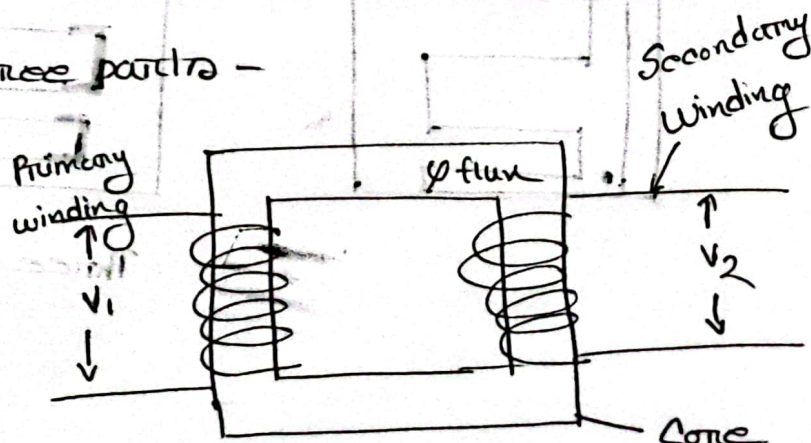
Adjust the voltages
frequency constant.



It increases the voltage at the power generating station (step up) and used to decrease the voltage (step down) for house and purpose.

Transformers have mainly three parts -

1. Core
2. Primary winding and
3. Secondary winding



The main problem with the transformer core is its hysteresis and eddy current losses. To reduce the losses the core is constructed with laminated steel sheet which

■ high silicon content

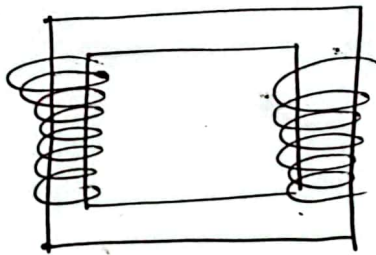
■ high permeability and

■ low hysteresis loss.

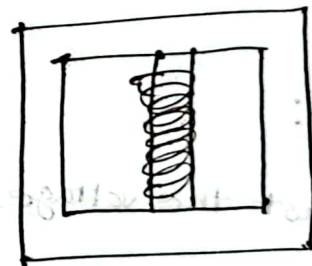
Core types

1. Core type

2. Shell type

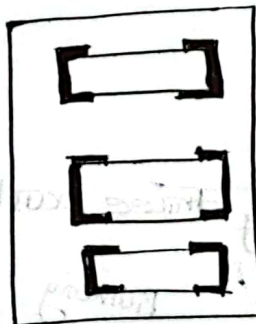
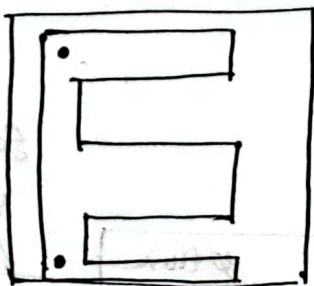


Core



Shell

Shell type

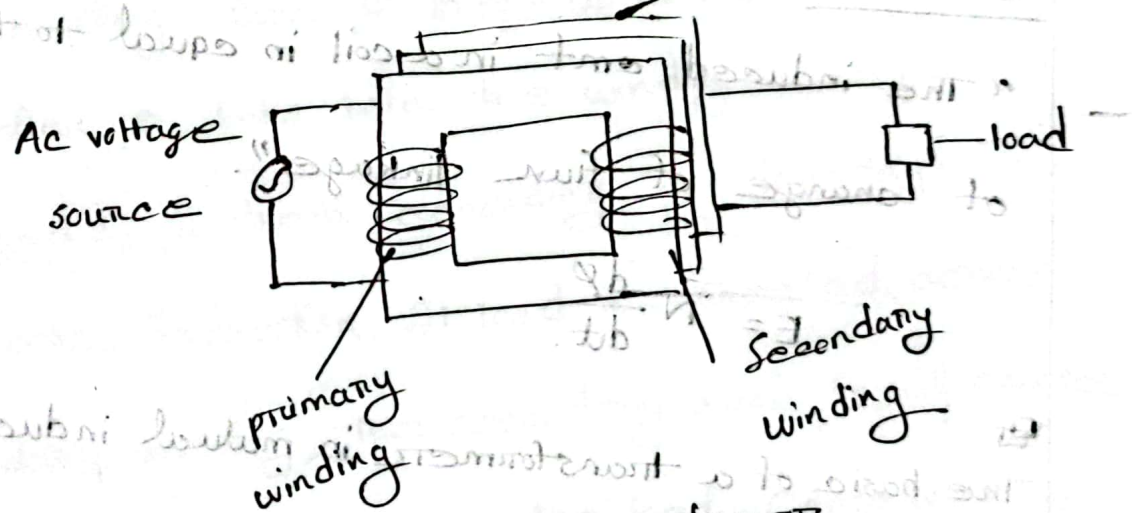


Three phase



Working Principle :-

It is essentially consist of two windings. The primary and secondary winding wound on a common laminated magnetic core. The winding connected to ac source is called primary winding. In secondary winding the core is connected.



Electrical transformer

Transformer works on the principle of Faraday's Electro-magnetic mutual Induction.

Faraday's first law :-

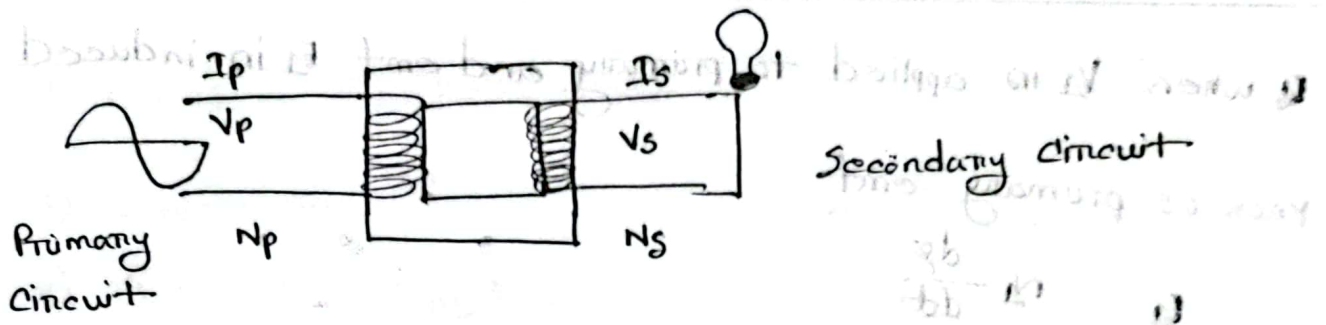
"Whenever a conductor is placed in a varying magnetic field, an electromotive force (emf/voltage) is induced. If the conductor circuit is closed, a current is induced which is called induced current".

2nd law :-

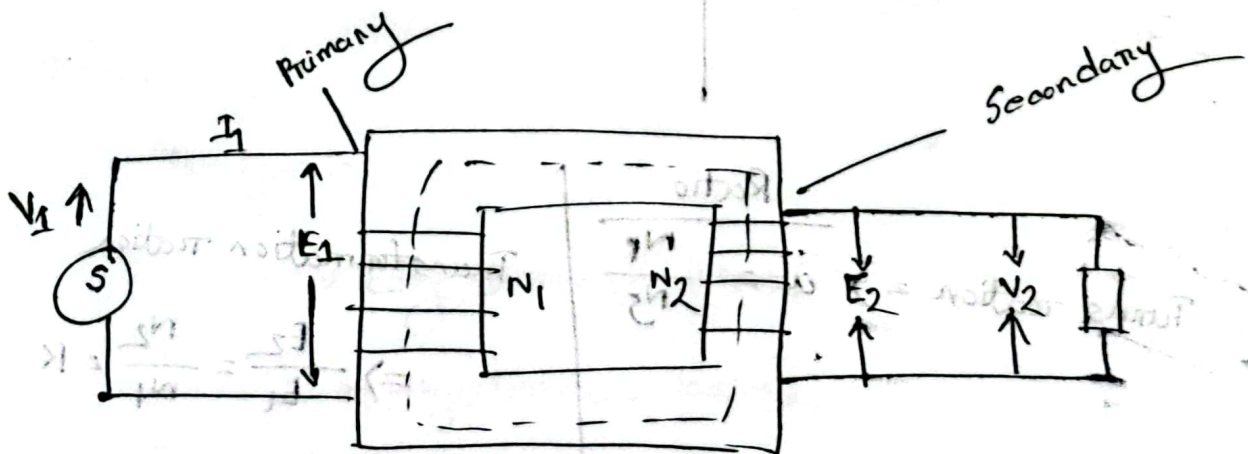
"The induced emf in a coil is equal to the rate of change of flux linkage".

$$E = N \frac{d\phi}{dt}$$

The basic of a transformer is mutual induction between two circuits linked by a common magnetic flux. If one coil of transformer is connected to a source of alternating voltage, an alternating flux is set up in the laminated core, most of the flux is linked with the other coil, according to Faraday's law of electromagnetic induction.



The alternating voltage V_1 is applied to the primary. Depending upon the number of turns of the primary (N_1) an alternating flux ϕ is set up in the core. This alternating flux ϕ links both the windings and induces emf's E_1 and E_2 in them according to Faraday's law of electromagnetic induction. If load is connected across the secondary winding, the secondary emf, E_2 will cause a current I_2 to flow through the load. Consequently terminal voltage V_2 will appear across the load.



Q when V_1 is applied to primary and emf E_1 is induced known as primary emf

$$E_1 = -N_1 \frac{d\phi}{dt}$$

$$E_2 = -N_2 \frac{d\phi}{dt} \quad (\text{secondary emf}).$$

$$\frac{E_2}{E_1} = \frac{N_2}{N_1}$$

step up

$$N_s > N_p$$

$$E_s > E_p$$

$$V_s > V_p$$

step-down

$$N_s < N_p$$

$$E_s < E_p$$

$$V_s < V_p$$

Turns ratio = $a = \frac{N_p}{N_s}$

Transformation ratio

$$\Rightarrow \frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$

Losses in Practical Transformers.

i) Copper loss

ii) Core loss / Iron loss

iii) Leakage flux.

Copper loss

The resistive heating losses in the primary secondary windings of transformer.

To determine the copper losses, short circuit test is performed.

$$\text{Copper loss} = I_1^2 R_1 + I_2^2 R_2$$

Core / Iron loss

These losses occur due to alternating flux set up in the core.

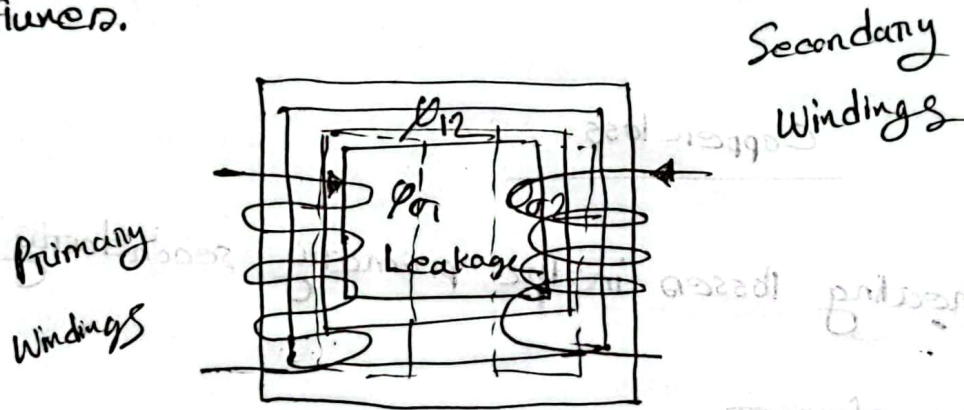
Two types :-

* Eddy current losses.

* Hysteresis losses.

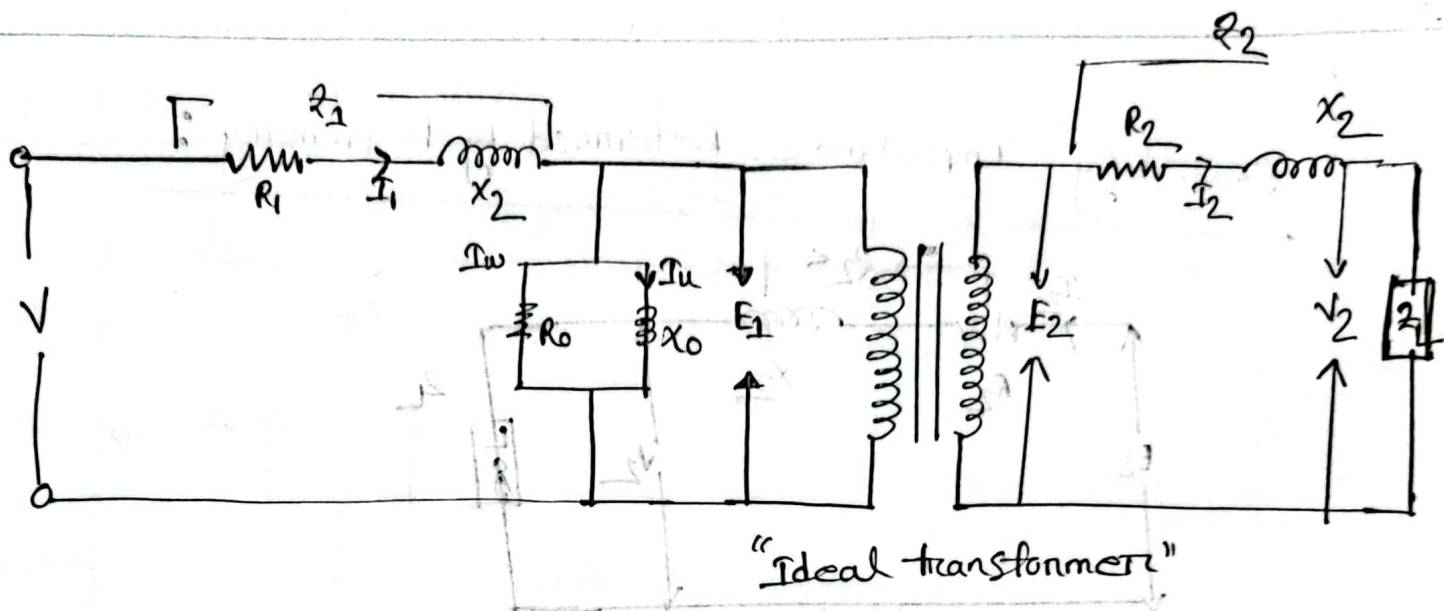
Leakage flux

The fluxes which escape the core and pass through only one of the transformer windings are leakage fluxes.



"Practical Transformer Equivalent Circuit"

=> Basically an equivalent circuit in which the resistance and leakage reactance of the transformer are imagined to be external to the windings.



Shifting Impedance by Turns Ratio:

$$Z_L = \frac{V_L}{I_L}$$

$$Z_L = \frac{V_S}{I_S}$$

The apparent impedance of the primary circuit:

$$Z_L' = \frac{V_P}{I_P} \quad Z_L' = \frac{V_P}{I_P} = \frac{a V_S}{I_S / a} = a^2 \frac{V_S}{I_S}$$

for secondary Imp* :-

note $\left[a = \frac{I_S}{I_P} / a = \frac{V_P}{V_S} \right]$

So,

$$R_2' = R_2 / k^2$$

$$X_2' = X_2 / k^2$$

$$Z_2' = Z_2 / k^2$$

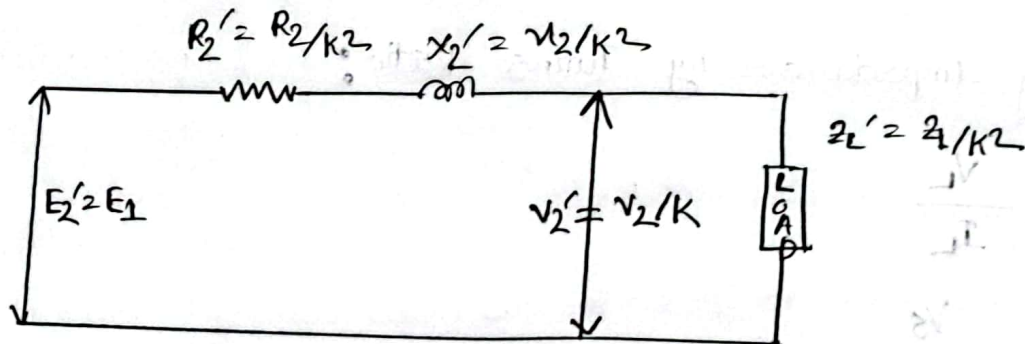
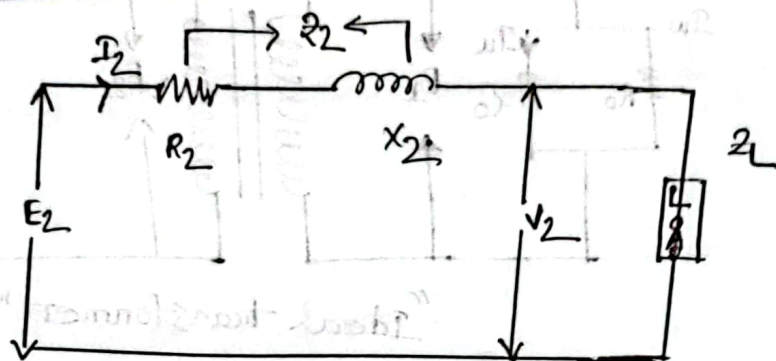
$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = k$$

$$\therefore E_2' = E_2 / k$$

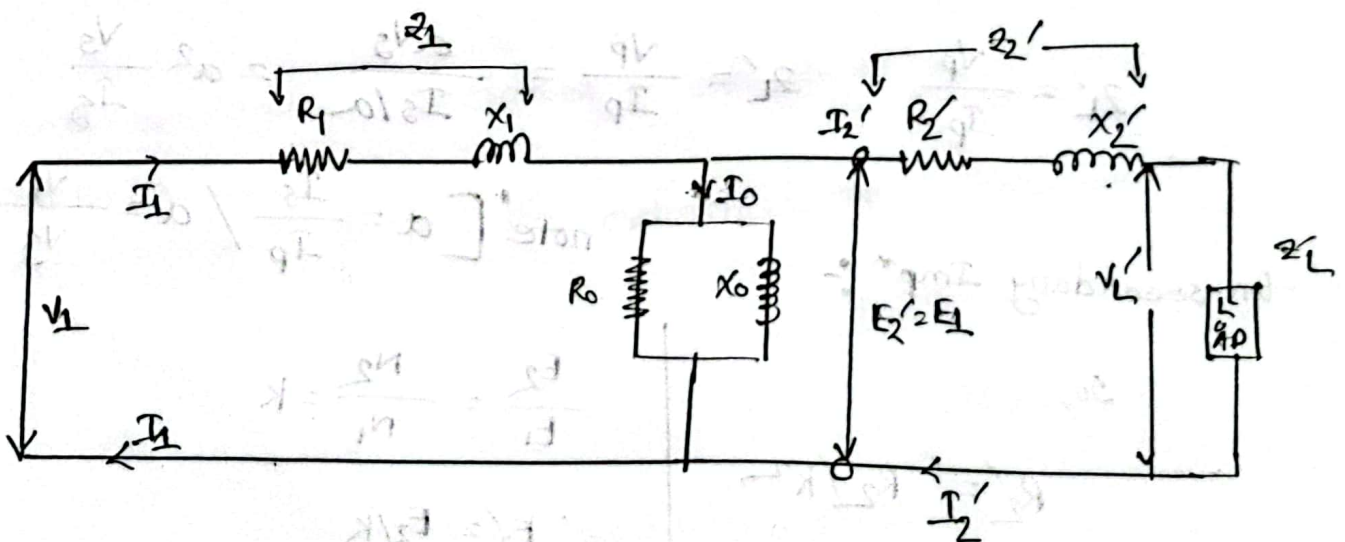
$$V_2' = V_2 / k$$

$$I_2' = I_2 k$$

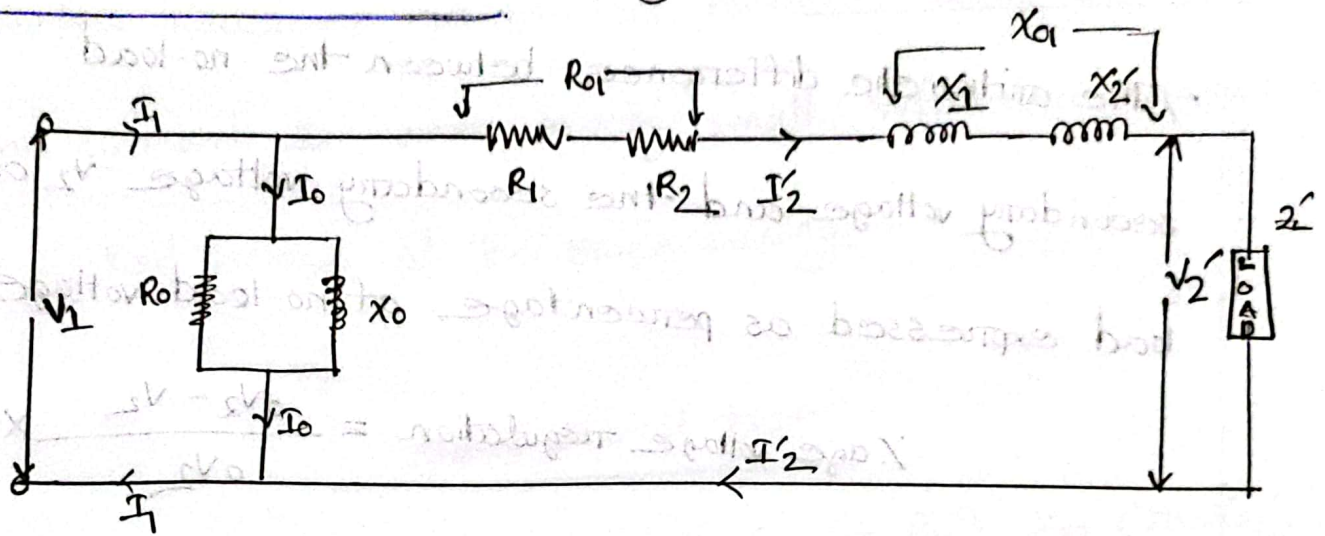
Secondary Impedance Referred by to Primary :



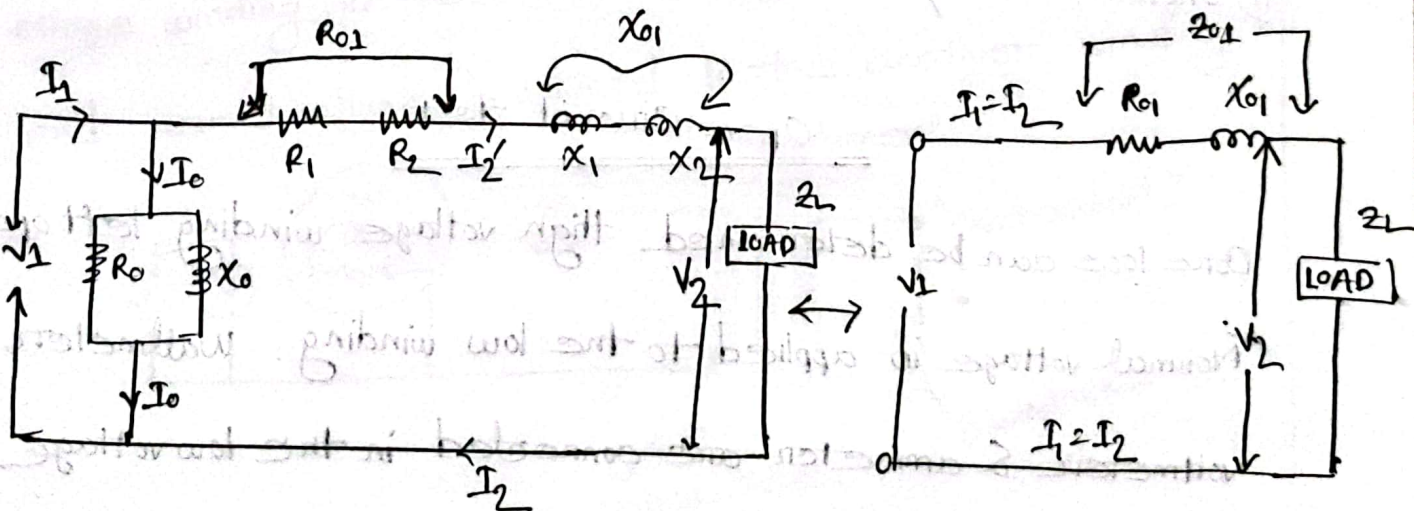
Now: Equivalent resistance



Exact Equivalent Circuit : of fig "01".

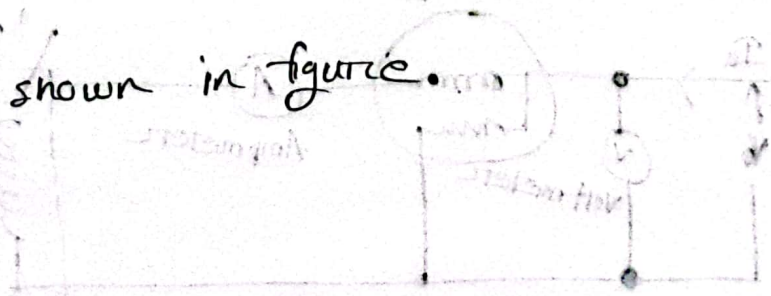


Approximate Equivalent Circuit



At last, the circuit is simplified by omitting I_0

altogether as shown in figure.



Voltage Regulation:

⇒ The arithmetic difference between the no-load secondary voltage and the secondary voltage V_2 on load expressed as percentage of no-load voltage.

$$\% \text{ age voltage regulation} = \frac{V_2 - V_L}{V_2} \times 100$$

Test

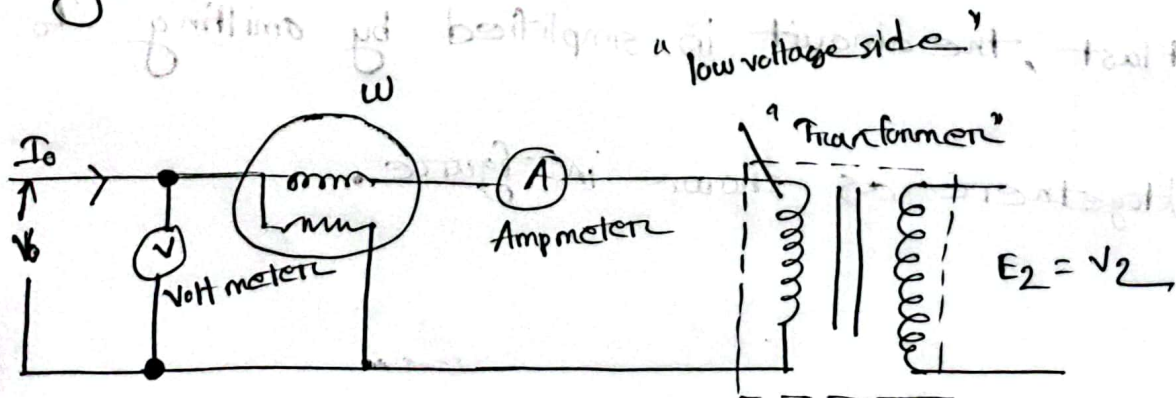
- i) Open circuit / No load test.
- ii) Short circuit / Impedance test.

Open Circuit test

Core loss can be determined. High voltage winding left open.

Normal voltage is applied to the low winding. Wattmeter, voltmeter & ammeter are connected in the low voltage

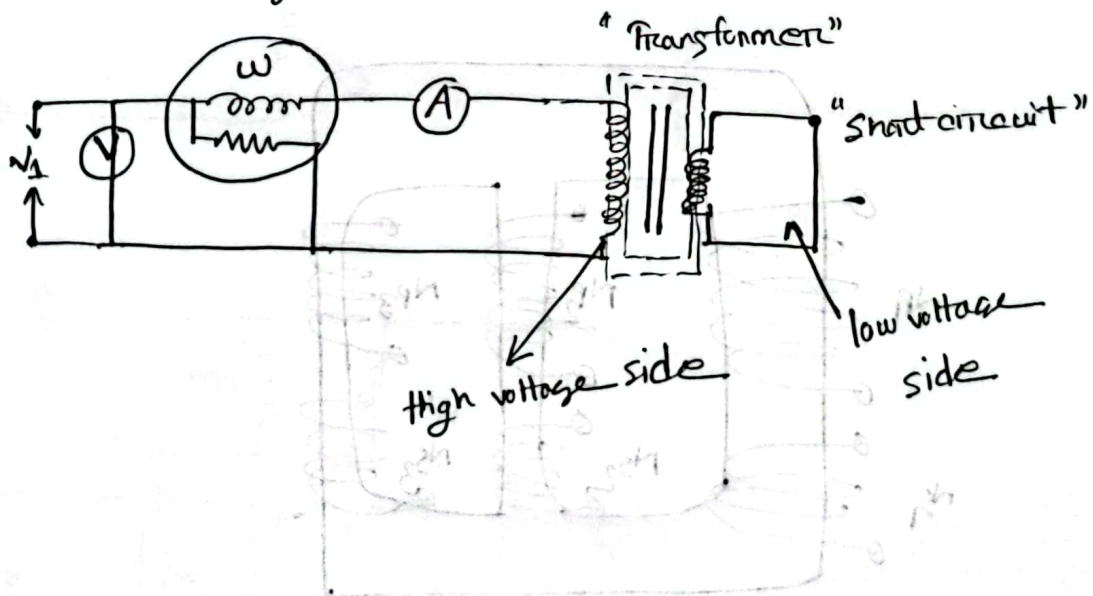
winding.



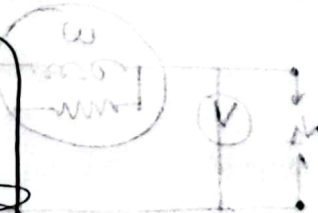
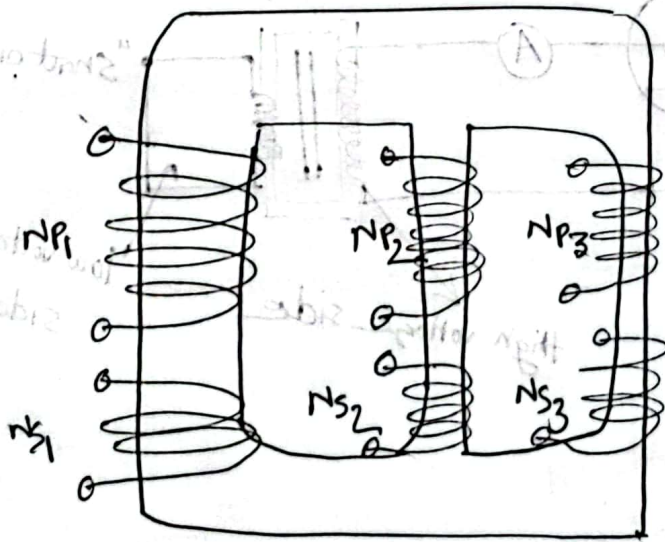
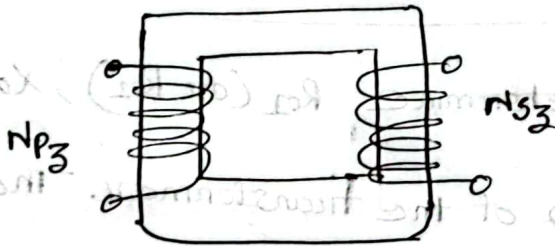
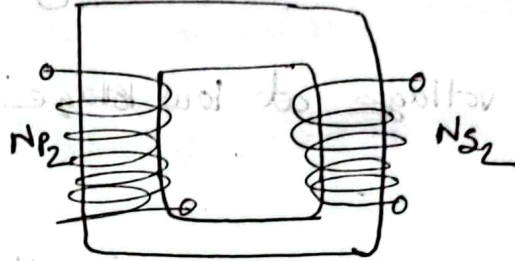
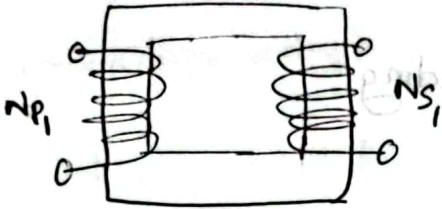
Wattmeter measures the iron loss. Ammeter measures the no load current I_0 which is very small. Voltmeter measures the applied voltage at low voltage winding.

Short Circuit test

This test is conducted to determine R_{01} (or R_{02}), X_{01} (or X_{02}) and full load copper losses of the transformer. The low voltage winding is short-circuited by a thick conductor and normal voltage is applied to the another winding.



Three phase Transformer



The four most common connection.

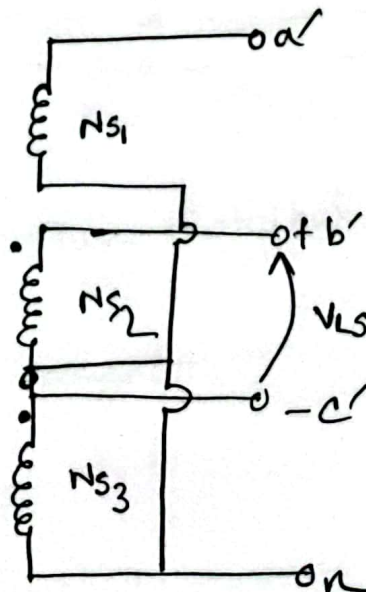
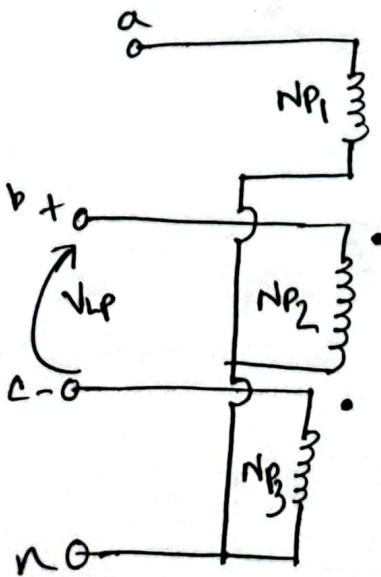
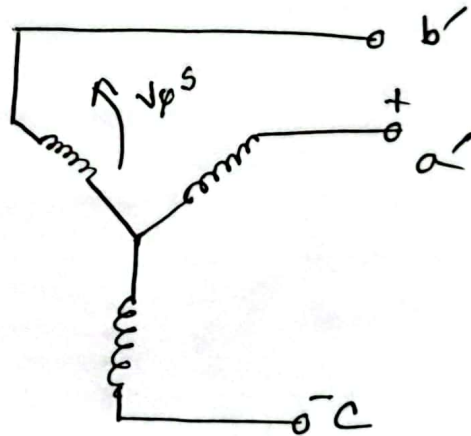
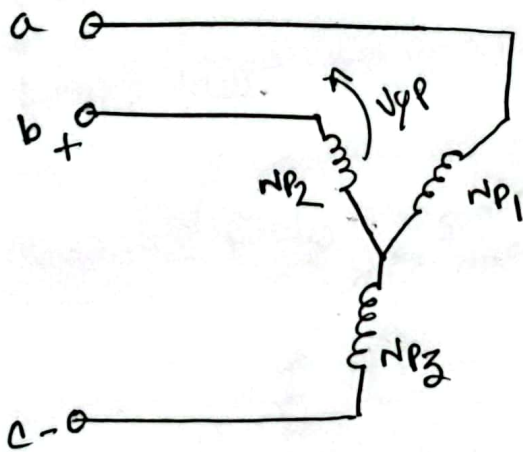
i) Y-Y

iv) D-Y

ii) D-D

iii) Y-D

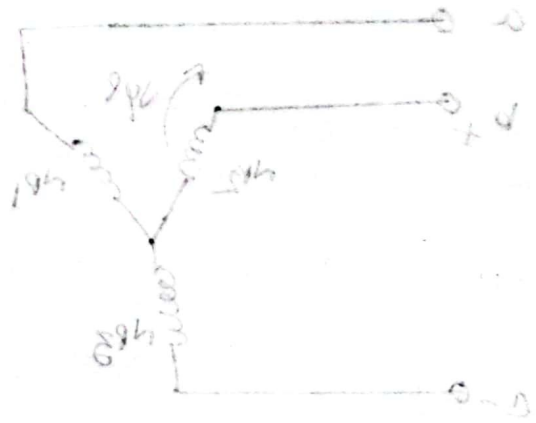
Y-Y - 3 ϕ connections



$$\frac{V_{LP}}{V_{LS}} = \frac{\sqrt{3} V_{pP}}{\sqrt{3} V_{pS}} = \alpha$$

Y-Y problem

Voltage become unbalanced when loads on the transformer circuit are unbalanced 3rd harmonic can be large



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Segment - 1

Electromagnetic Induction

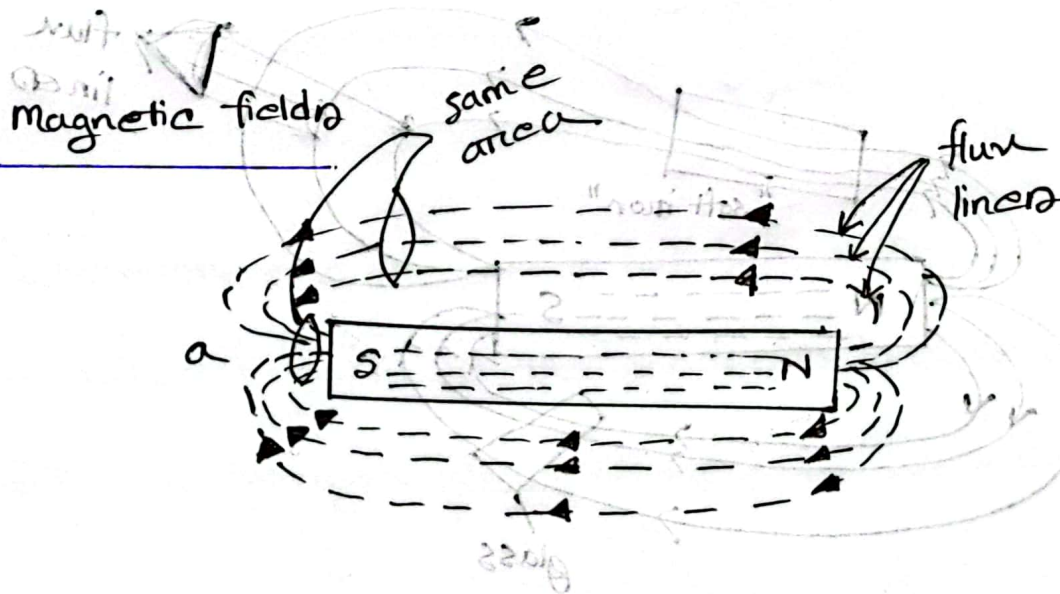
Magnetic field

Faraday's law of electromagnetic induction

Voltage induced in a conductor

Direction of induced voltage

Lenz's law.

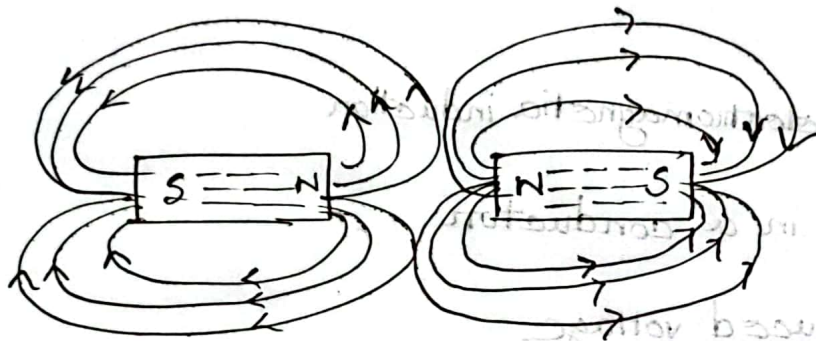


(a) Fig: Flux distribution for a permanent magnet.

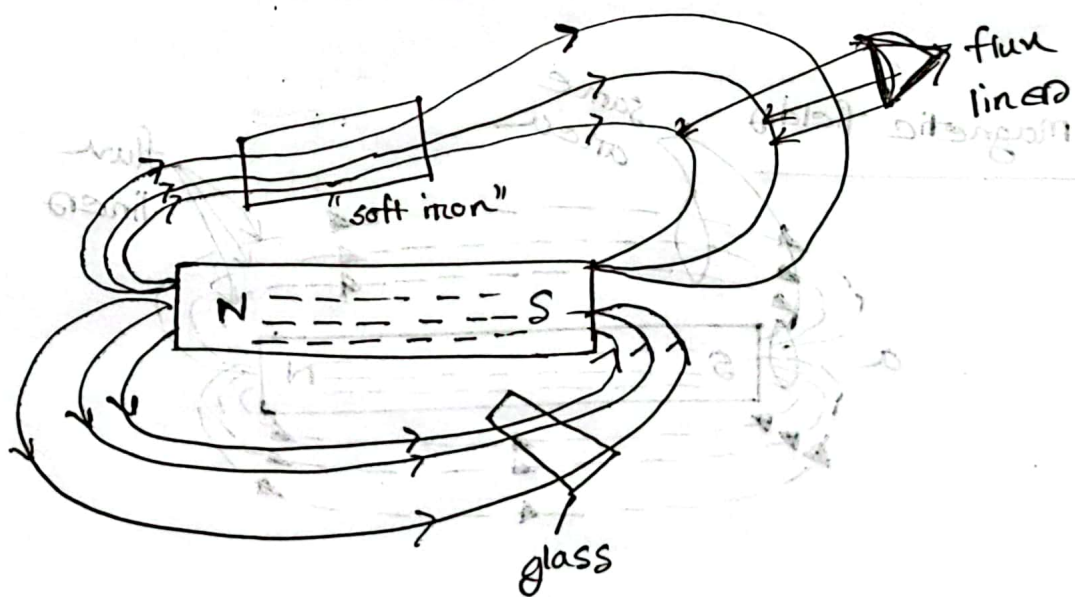
(b) Fig: Flux distribution for a permanent magnet.

Q The magnetic flux lines radiate from north to south pole, return to the north pole through the metallic bar.

Q strength depends on density of flux lines in that region.



If poles brought together, magnets will repel.



- i) If we place nonmagnetic material (copper/glass), there will be unnoticeable change in the flux distribution.
- ii) If we place magnetic material (soft iron), flux line will pass through it rather than air.

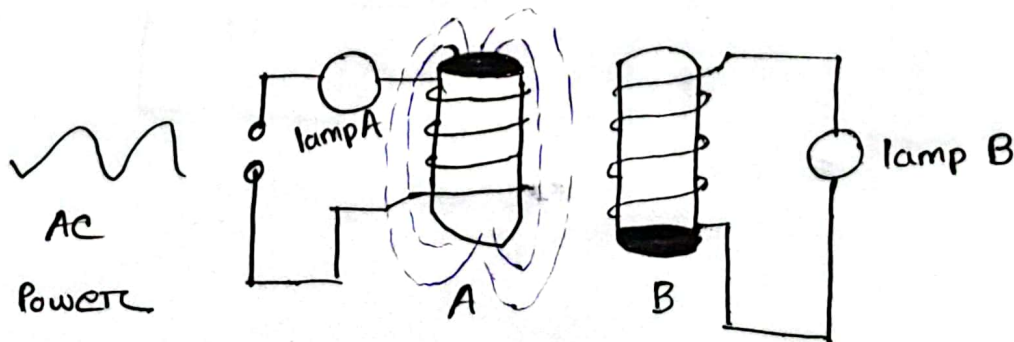
DC generator

Two Types $\begin{cases} \text{DC} \\ \text{AC} \end{cases}$

- Both convert mechanical power to electrical power.
- DC generator produce direct power, where AC does alternating power.
- Both work on (electrical power) Faraday's law of electromagnetic induction.

Faraday's first law:

"Whenever a conductor is rotated in magnetic field emf is induced in it. If the conductor is closed, current will flow through it which is known as induced current."

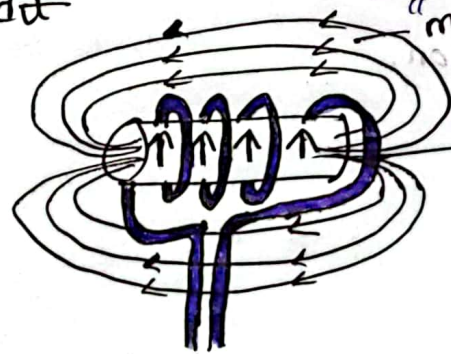


Faraday's 2nd law:

E.M.F is equal to the rate of change of linkage or flux cut.

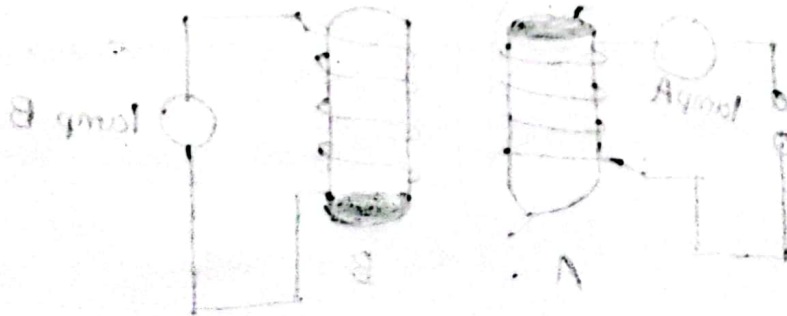
This EMF will cause a current to flow if the conductor circuit is closed.

$$\text{emf} = -N \frac{d\phi}{dt}$$



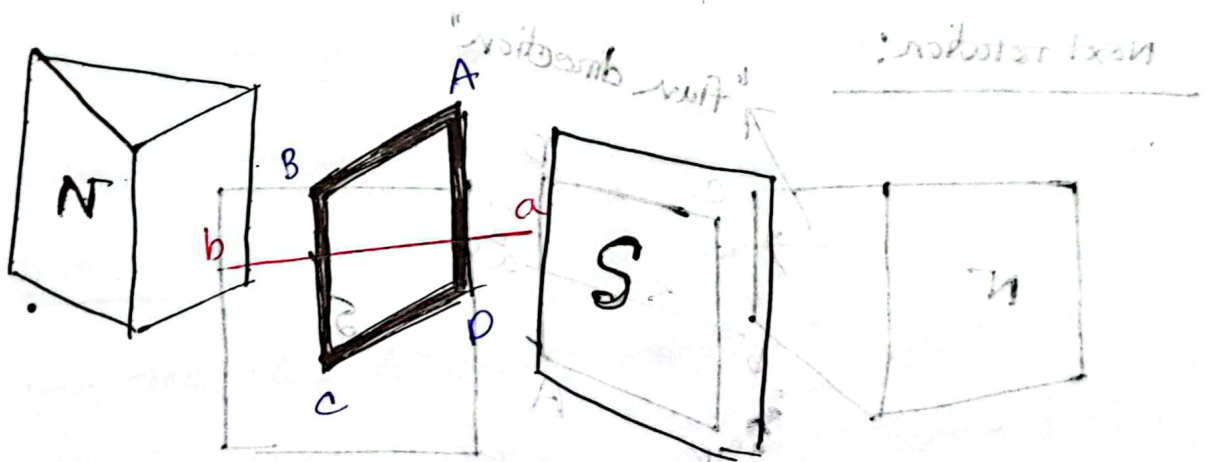
Q Fleming's right hand law.

Q " left hand law.



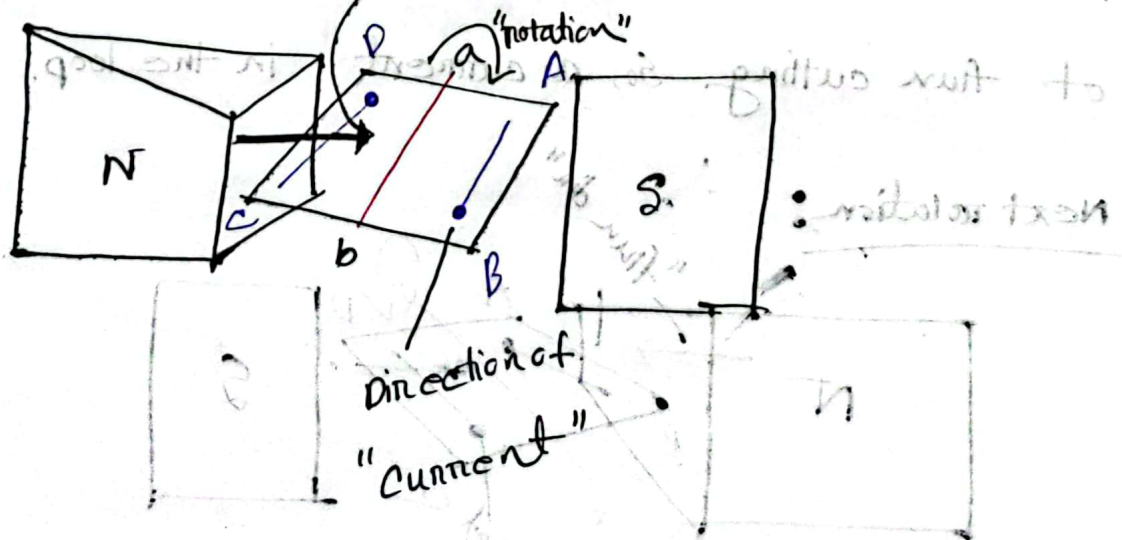
"Single loop DC generator"

It is a rectangular shape (single loop of conductor) is placed between two opposite poles of a magnet.



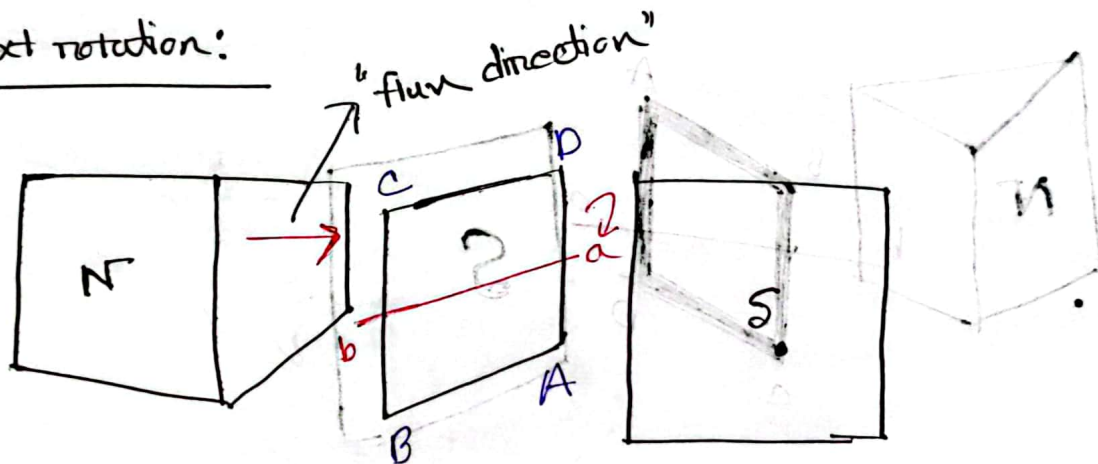
ABCD rotates inside the magnetic field axis ab.

"Direction of flux"



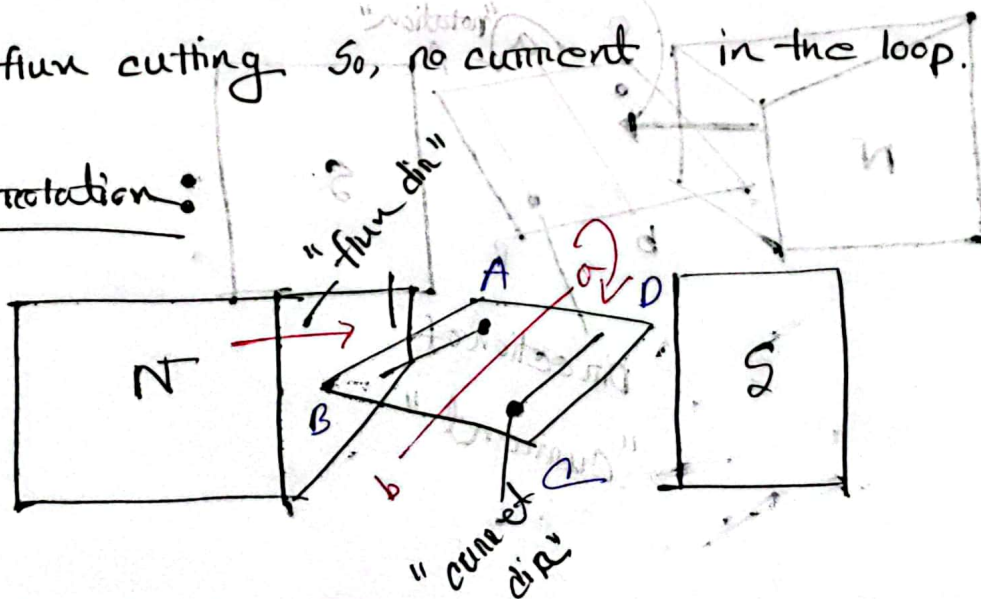
When loop rotates from its vertical position to its horizontal position, it cuts the flux line of the field EMF induced in both sides. If we apply Fleming's right hand rule, current $A \rightarrow B$ / $C \rightarrow D$.

Next rotation:



In this case CD upper side and AB lower. At this position the tangential motion of the sides of the loop is parallel to the flux lines of the field so, no change of flux cutting so, no current in the loop.

Next rotation:



Now, AB side front of N pole and CD comes front of S pole. Here the tangential motion of the side of the loop is perpendicular to the flux lines, hence the rate of flux cutting is maximum. Current flow $B \rightarrow A$.

$$\left[\frac{d\phi}{dt} = v \right] \quad \text{or} \quad \frac{d\phi}{dt} = v \times \frac{d\phi}{dx} = \frac{d\phi}{dt}$$

$D \rightarrow C$.

Discussion:

Every time the side AB comes in front of S pole, the current flows from $A \rightarrow B$. When it comes to N pole, current flows B to A.

Similarly CD in front of S pole, current direction C to D.

CD in front of N pole, current direction D to C.

$$E = \frac{d\phi}{dt} = \frac{d(B \cdot A)}{dt} = B \cdot \frac{dA}{dt}$$

Since $\frac{dA}{dt} = v \times \frac{dA}{dx} = \frac{dA}{dt}$

$$E = B \cdot \frac{dA}{dt} = B \cdot v \times \frac{dA}{dx} = B \cdot v \times \frac{dA}{dx}$$

$$E = B \cdot v \times \frac{dA}{dx}$$

Operation of DC Motor :

Let, $E = 0$

$$dF = dq \times v \times B$$

It is the cross product of dq , v and magnetic field B .

$$dF = dq \frac{dL}{dt} \times B \quad \left[v = \frac{dL}{dt} \right]$$

Where, dL is the length of the conductor carrying charge

q.

$$dF = dq \frac{dL}{dt} \times B$$

$$\text{or } dF = I dL \times B \quad \left[I = \frac{dq}{dt} \right]$$

$$\text{or } F = IL \times B = ILB \sin \theta$$

$$\therefore F = ILB \sin \theta$$

When force to left hand side,

$$F = ILB \sin 90^\circ = ILB$$

When force is on right-side

$$F = (-I)LB \sin 90^\circ$$

$$= -ILB$$

The torque produced is given by,

Torque = (force tangential to the direction of armature rotation) $\times$ distance

$$\text{or, } T = F_{\text{cond}} \times w$$

$$\text{or, } T = BILw \text{ cond}$$

Motor Duty

Classes of motor duty:

Continuous Duty with Intermittent Periodic loading
Starting and Braking

Periodic Speed changes

Motor de-rating:

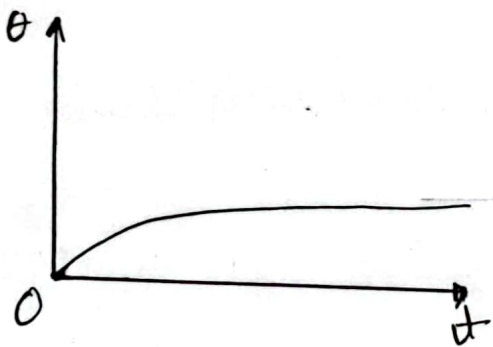
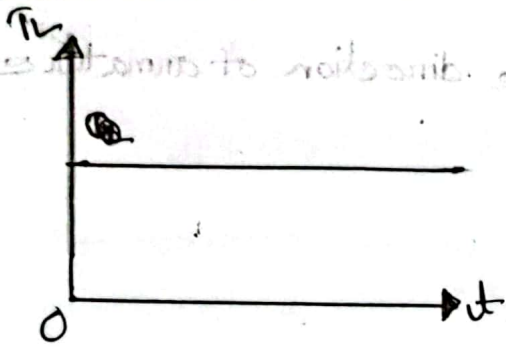
Ambient temperature

Mounting Arrangement

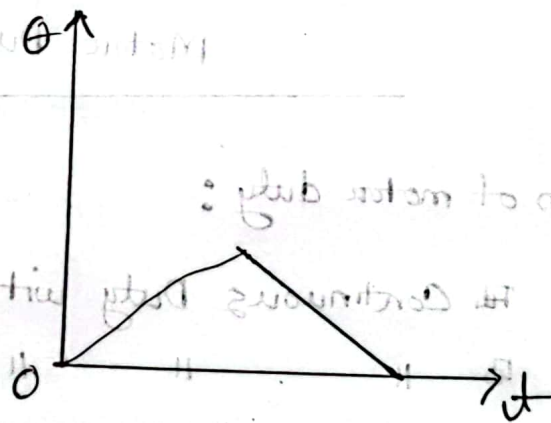
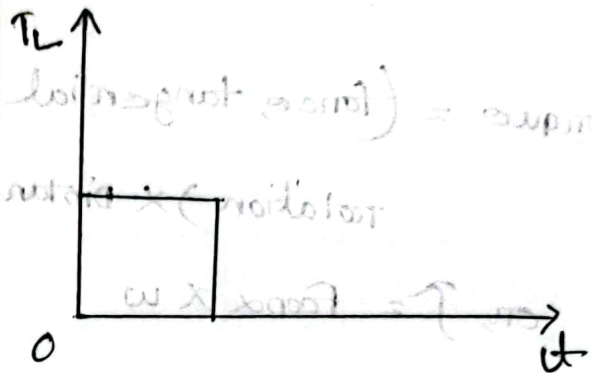
40°C

$$\text{Efficiency} = \frac{\text{Mechanical Power Output}}{\text{Electrical power Input}} \times 100\%$$

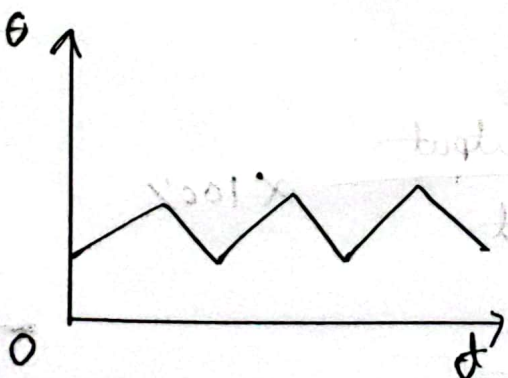
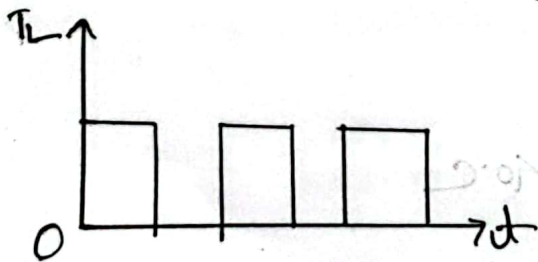
Continuous Duty Cycle :-



Short-time Duty cycle :-

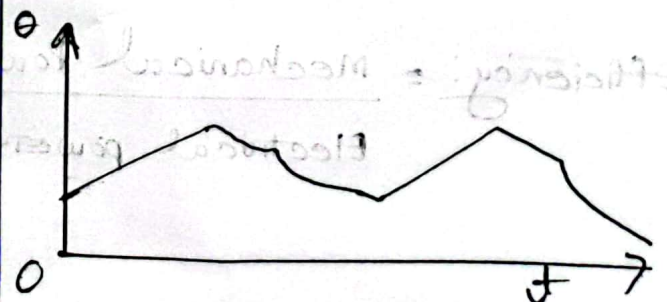
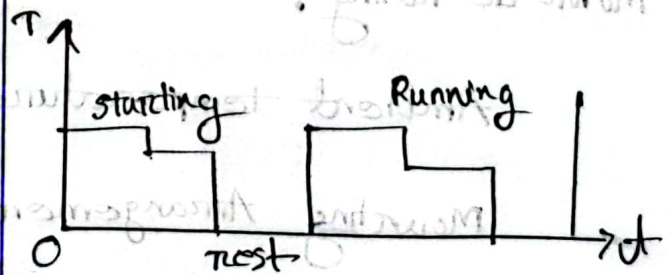


Intermittent Periodic Duty :-



Intermittent Periodic Duty with

starting:-



Motor Sizing

It converts electrical energy into mechanical energy

measured in Watts

1 hp = 746 W

Speed: The number of revolutions per minute.

denoted by 'N'

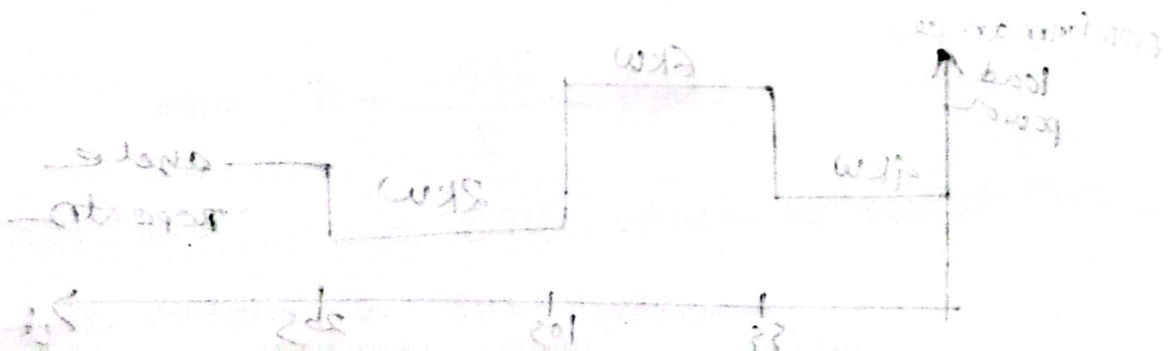
Torque: The force with which a motor turns is called

Torque measured in (Newton-meter) Nm

Relation between Power, Speed & torque

$$P = \frac{2\pi TN}{60}$$

$$\text{RMS power} = \sqrt{\frac{P_1^2 t_1 + P_2^2 t_2 + \dots + P_n^2 t_n}{t_1 + t_2 + \dots + t_n}}$$



Q

Can motor start the load?

Accelerating Torque = $T_{\text{motor}} - T_{\text{load}}$.

If $T_{\text{motor}} > T_{\text{load}}$. load will accelerate

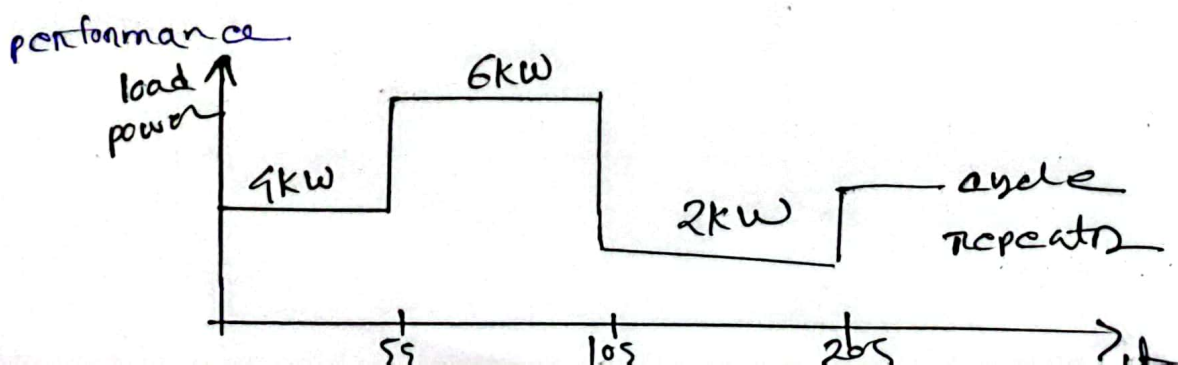
Else if $T_{\text{motor}} < T_{\text{load}}$. load will decelerate

Peak load torque (or power) < Rated motor torque

$\times (1 + \text{Service factor} / 100\%)$.

Example

The following load torque profile was measured for a constant speed load. choose an appropriately sized motor to drive this load assuming a service factor of 20%. You don't need to count starting



Peak load power requirement = 6kW.

Peak load must be less than Rated motor power $\times$ Service factor/100.

$$\Rightarrow 6\text{kW} < P_{\text{motor}} \left(1 + \frac{20\%}{100}\right)$$

$$\Rightarrow 6\text{kW} < P_{\text{motor}} (1.2).$$

$$\Rightarrow P_{\text{motor}} > 5\text{kW}$$

$$\text{OR } P_{\text{motor}} = 5\text{kW}.$$

The RMS load power should ideally be between 75% and 100% of rated motor.

$$\text{RMS load power} = \sqrt{\frac{P_1^2 t_1 + P_2^2 t_2 + P_3^2 t_3 + \dots + P_n^2 t_n}{t_1 + t_2 + \dots + t_n}}$$

$$= \sqrt{\frac{4^2 \times 5 + 6^2 \times 5 + 2^2 \times 10}{5 + 5 + 10}}$$

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$$= 4.16\text{kW}.$$

$$\text{Now, } K = \frac{4.16}{5} \times 100\%$$

$$= 83\% \text{ which is acceptable}$$

So, choose a 5kW motor.